

GARCH–Autoformer: Hybrid Residual Correction under Multi-Distribution Accuracy–Risk Evaluation

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Abstract. Volatility forecasting must preserve point accuracy while supporting calibrated tail-risk decisions. We propose GARCH–Autoformer, a hybrid residual-correction architecture that combines a structured GJR–GARCH variance path with a decomposition encoder operating on standardized innovations. Positive variance fusion retains the econometric baseline while allowing nonlinear multi-horizon corrections. Risk is reconstructed independently with Normal, Student- t , and filtered-historical-simulation VaR; method-level backtest criteria are averaged before SAW and TOPSIS ranking. The study contains 1,322,330 predictions from 26 configurations, nine equity markets, and five horizons. Under this primary multi-distribution definition, MoiraiVaR–Moirai-MoE ranks first under both 50:50 and 30:70 accuracy–risk policies. GARCH–Autoformer Tier 2 ranks twelfth under both policies, although it ranks first when risk is measured by Student- t VaR alone. The result identifies a distribution-specific strength rather than distribution-robust superiority and shows why a single VaR assumption can materially change architectural conclusions.

Keywords: GARCH–Autoformer; volatility forecasting; Value-at-Risk; hybrid architecture; multi-criteria decision making

1 Introduction

Volatility is latent, persistent, and operationally linked to market risk. ARCH and GARCH provide interpretable conditional-variance dynamics [1, 2]; the asymmetric GJR extension also represents leverage effects [3]. Deep sequence models can learn richer temporal patterns, but a lower point loss does not by itself imply realistic volatility amplitude or calibrated Value-at-Risk (VaR). Rankings also depend on the target, proxy, and loss [4]. A useful architecture should therefore improve flexibility without discarding the inductive structure needed for risk.

We introduce *GARCH–Autoformer*, a hybrid architecture designed around that requirement. GJR–GARCH supplies a recursive variance path. A decomposition encoder inspired by Autoformer receives standardized innovations and predicts a horizon-wise variance correction. The two branches are fused in variance space with a positive link. Training uses a Student- t likelihood, connecting the learned scale to heavy-tailed returns rather than only to a volatility proxy.

The paper makes three contributions. First, it specifies the new hybrid residual-correction design and two capacity tiers. Second, it evaluates the intended accuracy–risk balance with five forecast/dynamics criteria and four risk criteria averaged across three VaR constructions, rather than selecting on one error or one return distribution. Third, it isolates the scope of the architectural claim: GARCH–Autoformer is preferred under Student- t VaR alone, but not under the distribution-averaged primary analysis. This reversal is itself evidence that model-selection claims must report their VaR construction.

2 Related Work

GARCH remains a demanding benchmark even against richer econometric specifications [16]; FIGARCH extends this line to long-memory dynamics [17]. Neural hybrids later combined GARCH structure with recurrent networks [22], while Multi-Transformer–GARCH joined attention and econometric features [23]. Our design instead learns a signed, horizon-wise correction to a recursive GJR–GARCH path and fuses both branches directly in variance space.

The Transformer [18], Reformer [19], and Informer [20] provide relevant attention baselines. Autoformer introduced progressive decomposition [8]; FEDformer moved decomposition into Fourier/Wavelet representations [21]. Moirai and Moirai-MoE provide pretrained alternatives [10, 11]. Because simple linear models can outperform complex Transformers [9], our evaluation rejects degenerate low-dynamics forecasts before ranking.

Parametric Normal VaR is the classical variance–covariance baseline popularized by RiskMetrics [5]. Conditionally Student- t innovations represent heavier return tails [6], while filtered historical simulation (FHS) combines volatility filtering with an empirical innovation distribution [7]. We retain all three because each imposes a different tail shape; the primary risk score averages their diagnostics rather than allowing one distribution to determine the ranking.

SAW follows additive-utility aggregation [25], whereas TOPSIS uses distance from ideal and anti-ideal solutions [14, 26]. Reporting both exposes sensitivity to normalization and compensation.

3 Proposed GARCH–Autoformer

3.1 Structured and learned branches

For returns $r_t = \sigma_t z_t$, the GJR–GARCH branch estimates

$$\sigma_{G,t}^2 = \omega + \alpha \epsilon_{t-1}^2 + \gamma \mathbf{1}(\epsilon_{t-1} < 0) \epsilon_{t-1}^2 + \beta \sigma_{G,t-1}^2.$$

Parameters are fitted on the training split and recursively produce the base path $\sigma_{G,t+1:t+H}^2$. This branch gives persistence, asymmetry, and a stable scale prior.

The learned branch uses the last $L = 60$ standardized innovations $z_{t-L+1:t} = r/\sigma_G$. After within-window scaling, a linear embedding and positional encoding,

stacked decomposition blocks separate local trend and residual information. A feed-forward residual head maps the flattened representation to $\Delta_{1:H}$. Tier 1 uses $d = 32$, two encoder layers, and $d_{\text{ff}} = 128$; Tier 2 uses $d = 128$, three layers, and $d_{\text{ff}} = 512$.

3.2 Positive residual fusion and objective

As shown in Fig. 1, the hybrid output is

$$\hat{\sigma}_{t+h} = \sqrt{\text{softplus}(\sigma_{G,t+h}^2 + \Delta_h)}, \quad h = 1, \dots, H.$$

Predicting a correction instead of a complete volatility path anchors the network to an econometric baseline; allowing signed Δ_h lets it raise or lower that baseline. Softplus guarantees a positive final variance. With training-split degrees of freedom ν , parameters minimize the zero-mean Student- t negative log-likelihood

$$\mathcal{L} = \frac{1}{nH} \sum_{i,h} \left[\log \hat{\sigma}_{i,h} + \frac{\nu+1}{2} \log \left(1 + \frac{r_{i,h}^2}{\nu \hat{\sigma}_{i,h}^2} \right) \right].$$

This construction targets an accuracy–risk balance in two ways: nonlinear residual correction supplies forecast flexibility, while the GARCH path, positive scale, and heavy-tailed likelihood preserve a risk-compatible representation. The empirical protocol below tests whether that design goal is realized.

4 Accuracy–Risk Evaluation

4.1 Three VaR constructions

For $\alpha \in \{0.01, 0.05\}$, zero conditional mean, predicted scale $\hat{\sigma}_t$, Normal quantile z_α , and Student- t quantile $t_{\alpha,\nu}$, the first two left-tail thresholds are

$$\widehat{\text{VaR}}_{\alpha,t}^N = \hat{\sigma}_t z_\alpha, \quad \widehat{\text{VaR}}_{\alpha,t}^t = \hat{\sigma}_t t_{\alpha,\nu}.$$

Normal VaR follows the parametric RiskMetrics convention [5]. The Student- t degrees of freedom are fitted per market and bounded below by 2.1; the construction follows the heavy-tailed conditional model of Bollerslev [6]. For FHS, standardized residuals are $u_s = r_s / \hat{\sigma}_s$, and

$$\widehat{\text{VaR}}_{\alpha,t}^{FHS} = \hat{\sigma}_t \hat{Q}_\alpha \{u_{t-250}, \dots, u_{t-1}\}.$$

The one-step shift makes the threshold depend only on past residuals. This is the volatility-filtered empirical-quantile principle of Barone-Adesi et al. [7]. A violation occurs when $r_t < \widehat{\text{VaR}}_{\alpha,t}^m$.

Each method $m \in \{N, t, FHS\}$ is backtested separately in every market–horizon block. Let $PR_{\alpha,m}$ be the proportion of blocks passing both Kupiec

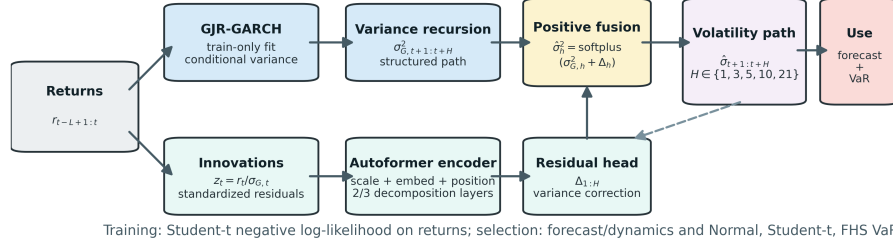


Fig. 1. GARCH–Autoformer. A structured GJR–GARCH variance path and a learned innovation correction meet in positive variance fusion.

coverage [12] and Christoffersen independence [13], and let $E_{\alpha,m} = |\bar{p}_{\alpha,m} - \alpha|$, where $\bar{p}_{\alpha,m}$ is the equally weighted mean block violation rate. The MCDM risk inputs are

$$\overline{PR}_{\alpha} = \frac{1}{3} \sum_m PR_{\alpha,m}, \quad \overline{E}_{\alpha} = \frac{1}{3} \sum_m E_{\alpha,m}.$$

Thus “VaR 1%” and “VaR 5%” in the primary ranking denote means of method-level diagnostics. The three numerical VaR thresholds are not averaged.

4.2 Forecast criteria and Sanity Gate

Let $y_i > 0$ and $\hat{y}_i > 0$ be observed and predicted volatility. Scoring rules must match the forecast target [24]; we use MSE, MAE, and QLIKE as complementary point losses, where

$$\text{QLIKE} = \frac{1}{n} \sum_i \left(\frac{y_i}{\hat{y}_i} - \log \frac{y_i}{\hat{y}_i} - 1 \right).$$

Dynamics are represented by amplitude error $e_{\sigma} = |s_{\hat{y}}/s_y - 1|$ and tracking error $e_{\rho} = 1 - \max(\rho(y, \hat{y}), 0)$. Together with $\overline{PR}_{.01}$, $\overline{E}_{.01}$, $\overline{PR}_{.05}$, and $\overline{E}_{.05}$, the decision matrix contains nine criteria.

A configuration is eligible only when every criterion is finite, $e_{\sigma} \leq 0.9$, and mean tracking correlation is positive. This Sanity Gate prevents a nearly constant forecast from compensating for missing dynamics through unrelated risk criteria.

4.3 Decision policies and statistical analysis

Simple Additive Weighting (SAW) [25] applies benefit/cost min–max normalization and sums weighted scores. TOPSIS uses vector normalization and ranks closeness to the weighted ideal [14, 26]. Their ordinal ranks are averaged; lower consensus rank is better. The balanced policy allocates 50% to forecast/dynamics and 50% to VaR. The risk-priority policy allocates 30:70. Criteria receive equal

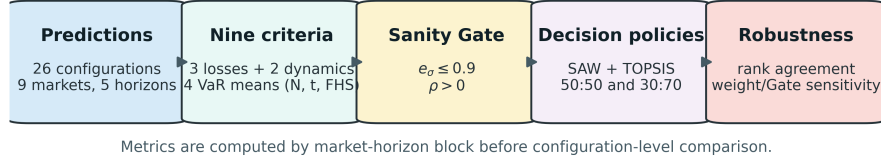


Fig. 2. Experimental evaluation under explicit accuracy–risk preferences. Each VaR criterion first averages Normal, Student- t , and FHS diagnostics.

weight within each group; method averaging occurs before SAW/TOPSIS normalization.

Figure 2 summarizes the evaluation. Metrics are first computed for each of 45 market–horizon blocks and then aggregated by configuration. Forecast-loss differences are assessed with the Friedman test and Kendall’s W , following block-wise comparison guidance [15]. Rank agreement, weight sweeps, and Gate sweeps test whether the conclusion depends on one decision setting.

5 Experimental Study

5.1 Data and competitors

The experiment contains 1,322,330 daily predictions for VN-Index, VN30, KOSPI, Nikkei 225, DAX 40, Euronext 100, IBEX 35, SMI, and S&P 500 at horizons $\{1, 3, 5, 10, 21\}$. The 26 configurations include four GARCH variants, including FI-GARCH [17] and GARCH–LSTM [22]; twelve Autoformer, Informer, Reformer, and vanilla-Transformer tiers [8, 20, 19, 18]; three Moirai-family models [10, 11]; three VaR-aware variants; and four GARCH-/Wavelet–Autoformer hybrids. Autoformer is included because its decomposition principle is directly related to the learned branch [8]; linear and foundation-model results in prior work motivate heterogeneous comparators [9, 10].

5.2 Implementation details

Splits are chronological and market specific. Within GARCH–Autoformer, GJR–GARCH parameters and likelihood degrees of freedom are estimated on training observations. The post-hoc Student- t VaR evaluator fits one degree-of-freedom parameter per market; FHS uses a shifted 250-observation window. Both tiers use a 60-day input and produce 21 steps; reported horizons select steps 1, 3, 5, 10, and 21. Training uses AdamW (10^{-3} , weight decay 10^{-4}), batch size 128, at most 30 epochs, gradient clipping at 1, and validation-likelihood early stopping with patience 5.

5.3 Main rankings

The Gate retains 21 configurations and excludes five for excessive amplitude error, non-positive tracking, or both. Figure 3 shows the leading ranks.

MoiraiVaR–Moirai–MoE ranks first by both SAW and TOPSIS under 50:50 weights and again under 30:70 risk priority. GARCH–Autoformer Tier 1 and Tier 2 have consensus ranks 11 and 12 under 50:50; under 30:70, Tier 2 ranks 12 and Tier 1 ranks 15. The balanced ranks remain ahead of the corresponding plain Autoformer tiers (13 and 15), which is consistent with a useful hybrid correction within that model family, but not with overall superiority.

The winning configuration reports MSE 0.0253, MAE 0.1015, correlation 0.899, averaged pass rates 0.304 and 0.393, and averaged violation errors 0.00786 and 0.01008 at 1% and 5%. GARCH–Autoformer Tier 2 reports MSE 0.1690, MAE 0.2911, correlation 0.530, pass rates 0.304 and 0.230, and errors 0.01073 and 0.01207. Its 1% average pass rate is competitive, but its broader forecast and 5% risk profile is weaker.

Forecast losses differ across the 26 configurations: $\chi^2_{25} = 813.10$, 840.80, and 878.79 for MSE, MAE, and QLIKE (all $p < 10^{-150}$). Kendall’s W is 0.723, 0.747, and 0.781, respectively. Every method-level VaR Friedman test is also significant at $p < 0.001$. These omnibus statistics establish broad differences, not pairwise causal superiority.

5.4 Robustness of the balance claim

SAW and TOPSIS agree closely at 50:50 ($\rho_s = 0.981$, $p < 10^{-14}$) and remain high under 30:70 ($\rho_s = 0.877$, $p < 10^{-6}$). MoiraiVaR–Moirai–MoE remains first when the forecast/dynamics weight ranges from 30% to 70%. It also remains first as the Gate correlation threshold rises from 0 to 0.05 and 0.10, leaving 21, 16, and 13 configurations.

The conclusion is less stable across tail models. With Normal-only risk criteria, FI-GARCH is the consensus leader. With Student- t -only criteria, GARCH–Autoformer Tier 2 ranks first by both SAW and TOPSIS under both policies. Under FHS alone, the balanced leader is tied, while GJR–GARCH leads at 30:70. For GARCH–Autoformer Tier 2 specifically, the 1% block pass rates are 0.000, 0.422, and 0.489 for Normal, Student- t , and FHS. Averaging the three exposes the failure hidden by Student- t -only selection.

6 Discussion and Limitations

Four limitations bound the architectural interpretation. First, training protocols and observation coverage differ across competitors, so MCDM ranking is not a controlled ablation. Second, equal averaging of three VaR diagnostics is transparent but normative; FHS also has a shorter effective sample because of its 250-observation warm-up. Third, post-hoc Student- t degrees of freedom are estimated from the available per-market evaluation input rather than frozen

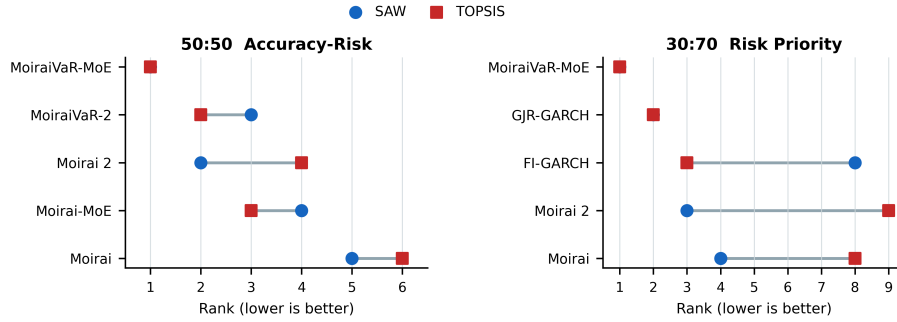


Fig. 3. Top five eligible configurations under balanced and risk-priority policies using three-method averaged VaR criteria. Lines expose SAW–TOPSIS disagreement.

from a training split, and all parametric VaR uses zero conditional mean. Fourth, code inspection found that the evaluated residual encoder selects autocorrelation delays but applies a fixed shift, and its nominal head-count parameter is unused. Consequently, the experiment does not establish that a faithful Autoformer autocorrelation operator caused an improvement. Correcting these issues and rerunning a purged, controlled ablation are required for a definitive architectural claim.

7 Conclusion

GARCH–Autoformer combines an asymmetric econometric variance path with a neural correction of standardized innovations and positive variance-space fusion. It improves on corresponding Autoformer tiers under the balanced ranking, and Tier 2 is the overall winner when the risk definition is restricted to Student- t VaR. It is not the distribution-robust winner: averaging Normal, Student- t , and FHS diagnostics places Tier 2 twelfth and selects MoiraiVaR–Moirai-MoE. The defensible conclusion is therefore that the proposed architecture has a Student- t -specific accuracy–risk strength, not a universal balance advantage. Future work will correct the delay operator, freeze all tail parameters on training data, add GARCH-only and neural-only ablations under identical splits, and test weighted or worst-case aggregation across VaR constructions.

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